FOUNDATION PROGRAMME / AGES 9–12

Young Problem Solvers

My Practice Workbook — Volume 1

Unit 1 · Think with Numbers | Weeks 1–3 | Six 90-minute sessions

Pilot Version 1.0 · Intended for educator review and testing; not an accredited syllabus or validated intervention.

Student name: _____ Group: _____

Use this workbook in class and for approved home practice. Show working, circle units, and write at least one sentence explaining important decisions. No speed rankings.

My checklist

- ☐ I understood the question.
- ☐ I drew, listed or calculated.
- ☐ I checked whether the answer makes sense.
- ☐ I explained a decision.

WEEK 1 · SESSION 1

Number Detectives: Place Value

Date: _____ Learning partner (optional): _____

Start here

How does the position of a digit change its value?

Independent practice — show your steps

1. Write 40,305 in expanded form.

2. In 57,219, what is the value of digit 7?

3. Use <, > or =: 28,409 __ 28,490.

4. Order from least to greatest: 9,200; 8,950; 9,020; 8,905.

5. A receipt shows 16,700 but a learner records 16,070. Describe the place-value error.

6. Write any two amounts between 12,450 and 12,500; explain why they fit.

Stretch challenge (optional)

A bookshop displays three prices: UGX 24,650, 24,560 and 25,046. A customer believes 24,560 is the highest because it ends in 560. Show why this is incorrect.

Exit ticket

Which is greater: 30,099 or 29,999? Explain using one place value.

Reflect

One answer I checked: _____

One idea I can now explain: _____

One question I still have: _____

My working space / second method

Use a number line, table, picture, or second calculation to prove one of your answers.

Instructor feedback (one strength, one next step): _____

WEEK 1 · SESSION 2

Estimate, Calculate, Check

Date: _____ Learning partner (optional): _____

Start here

How can an estimate help us catch mistakes?

Independent practice — show your steps

1. Round 6,450 to nearest 100 and nearest 1,000.

2. Estimate 2,780 + 5,120 to nearest 1,000.

3. Find the exact sum for Question 2.

4. Estimate and calculate 12,490 + 3,710. Show your chosen rounding.

5. A sign says 5,750 + 2,250 = 9,000. Correct it and explain.

6. Explain when rounding to 100 is more useful than rounding to 1,000.

Stretch challenge (optional)

A class has UGX 20,000 for two items costing UGX 9,650 and UGX 10,250. Will the money be enough? Estimate first and compute exactly.

Exit ticket

A total is reported as UGX 14,600 for 8,350 + 5,250. Is it right? Show a check.

Reflect

One answer I checked: _____

One idea I can now explain: _____

One question I still have: _____

My working space / second method

Use a number line, table, picture, or second calculation to prove one of your answers.

Instructor feedback (one strength, one next step): _____

WEEK 2 · SESSION 1

Budgets and Priorities

Date: _____ Learning partner (optional): _____

Start here

What must we pay for before we choose extras?

Independent practice — show your steps

1. From UGX 25,000, reserve UGX 3,000. What is left for supplies?

2. Two notebooks at UGX 2,500 each cost how much?

3. Three pencils at UGX 1,000 each cost how much?

4. Find the full required supplies cost: notebooks, pencils, tape UGX 4,000, poster UGX 2,000.

5. How much may be spent on extras while preserving the reserve?

6. Explain the difference between reserve and change after shopping.

Stretch challenge (optional)

You must also keep UGX 2,000 unspent IN ADDITION to the UGX 3,000 reserve. What is the new maximum amount available for optional items?

Exit ticket

Why is it misleading to say that UGX 11,000 is available for extras after buying the required supplies?

Reflect

One answer I checked: _____

One idea I can now explain: _____

One question I still have: _____

My working space / second method

Use a number line, table, picture, or second calculation to prove one of your answers.

Instructor feedback (one strength, one next step): _____

WEEK 2 · SESSION 2

Choices, Trade-offs and New Information

Date: _____ Learning partner (optional): _____

Start here

Which affordable plan best meets a goal?

_____

Independent practice — show your steps

1. Plan A: two markers at UGX 3,500. Find optional cost.

_____

_____

2. Plan B: marker UGX 3,500 plus poster UGX 2,000. Find optional cost.

_____

_____

3. With UGX 8,000 available, how much remains under Plan A and B?

_____

_____

4. Can you buy two markers AND a poster? Prove the answer.

_____

_____

5. A marker price rises to UGX 4,500. Recalculate Plan B.

_____

_____

6. Write a recommendation for either Plan A or B with two reasons.

_____

_____

Stretch challenge (optional)

New rule: keep at least UGX 2,000 unspent within the extras allowance. Can Plan A still be chosen? Make a revised recommendation.

_____

_____

_____

Exit ticket

Does “affordable” always mean “best”? Answer with one calculation-based example.

_____

_____

Reflect

One answer I checked: _____

One idea I can now explain: _____

One question I still have: _____

My working space / second method

Use a number line, table, picture, or second calculation to prove one of your answers.

Instructor feedback (one strength, one next step): _____

WEEK 3 · SESSION 1

Equal Groups and Whole Packets

Date: _____ Learning partner (optional): _____

Start here

What does a remainder mean when buying whole packets?

Independent practice — show your steps

1. Show 4 groups of 6 as repeated addition and multiplication.

2. 18 ÷ 4 equals how many full groups and what remainder?

3. How many 4-cup packets for 18 visitors?

4. How many cups are unused after buying those packets?

5. For 21 visitors, how many 5-cup packets are needed?

6. Explain why the answer “4.5 packets” is unsuitable for buying packets.

Stretch challenge (optional)

A packet contains 8 cups. How many packets serve 45 visitors, and how many unused cups remain?

Exit ticket

17 visitors, packets of 6 cups. How many packets and how many unused cups?

Reflect

One answer I checked: _____

One idea I can now explain: _____

One question I still have: _____

My working space / second method

Use a number line, table, picture, or second calculation to prove one of your answers.

Instructor feedback (one strength, one next step): _____

WEEK 3 · SESSION 2

Comparing Value for Money

Date: _____ Learning partner (optional): _____

Start here

What is the cheapest feasible way to supply enough cups?

Independent practice — show your steps

1. For 18 visitors, how many A packets (4 cups each) are needed?

2. What is the total A cost at UGX 2,000 each?

3. For 18 visitors, how many B packets (6 cups each) are needed?

4. What is the total B cost at UGX 2,700 each?

5. Which plan is cheaper and by how much?

6. Explain why choosing A just because one packet costs less can be misleading.

Stretch challenge (optional)

For 19 visitors, compare A and B again. Does the winner change? Explain.

Exit ticket

For 12 visitors, A (4 cups @ 2,000) versus B (6 cups @ 2,700): which costs less?

Reflect

One answer I checked: _____

One idea I can now explain: _____

One question I still have: _____

My working space / second method

Use a number line, table, picture, or second calculation to prove one of your answers.

Instructor feedback (one strength, one next step): _____

My learning portfolio: Unit 1

One calculation I can now explain and check:

A budget decision I made and why it was sensible:

A problem involving packets or groups that I solved:

Something I changed after receiving feedback:

My goal for Unit 2: